

STAT 525

EXAM I

STAT 525

Name _____

Problem	Points	Score
1	35	
2	45	
3	20	

USE YOUR TIME WISELY

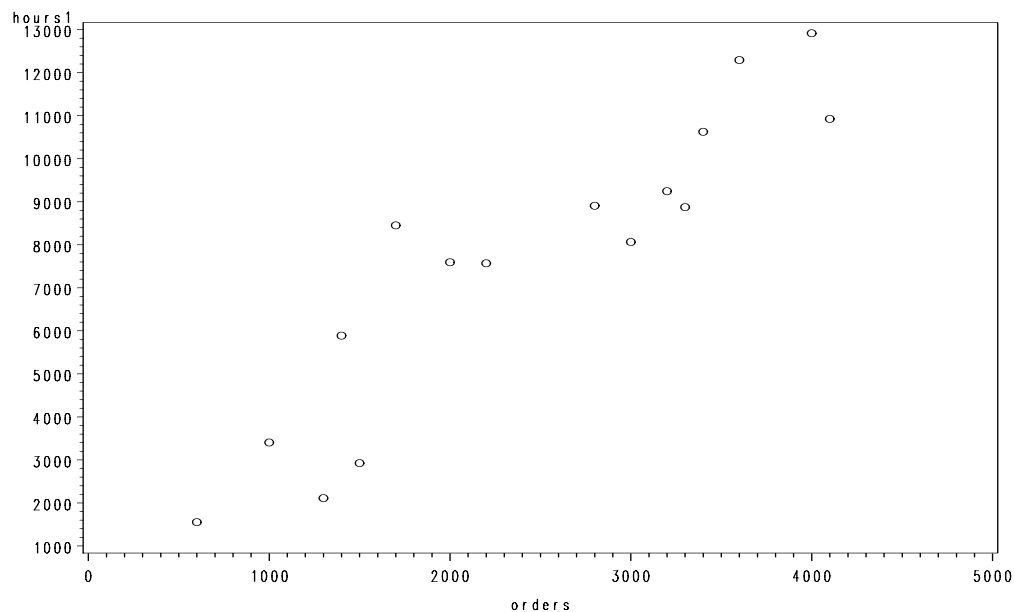
GIVE CLEAR, CONCISE ANSWERS

SHOW YOUR WORK TO RECEIVE PARTIAL CREDIT

WRITE LEGIBLY. ANYTHING UNREADABLE WILL NOT BE GRADED

GOOD LUCK!!!!

1. A manufacturing company is interested in the relationship between the number of orders processed per month (ORDERS) and the total work hours (HOURS1) needed to handle them. Use the attached SAS output at the end of the exam and the scatterplot below to answer the following questions.



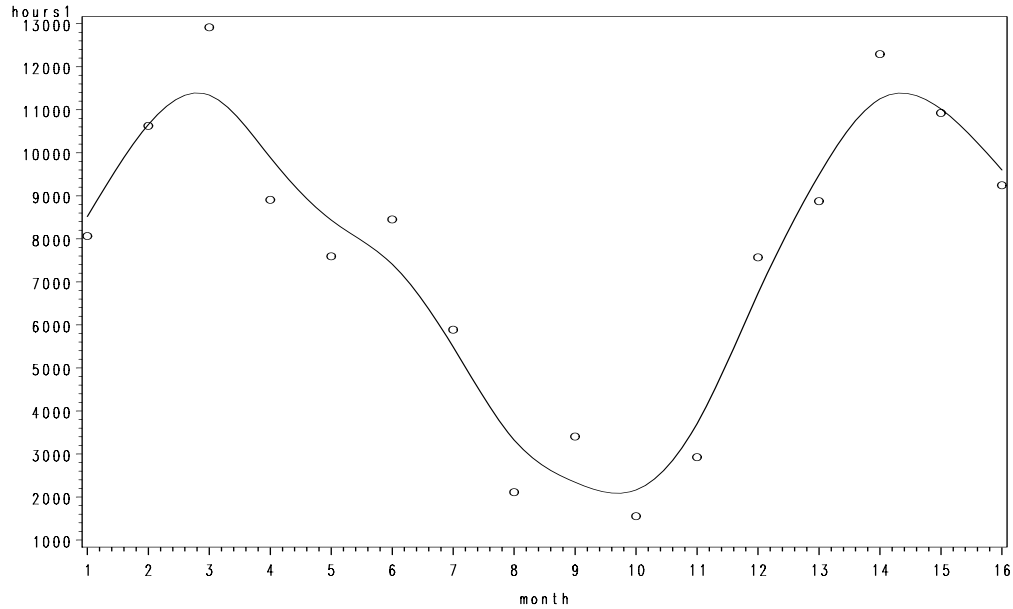
- (a) (5 pts) Write down the least squares regression line and describe the relationship between HOURS1 and ORDERS in terms that the CEO of this company would understand.

- (b) (4 pts) Suppose the CEO immediately dismisses this analysis saying that with zero orders, the model should predict zero hours. Describe how you would defend this analysis rather than resorting to refitting the data without an intercept.
- (c) (3 pts) One of the observations used to fit this model was (3000, 8065). What is the residual value for this observation.
- (d) (5 pts) Construct a 90% confidence interval for the slope and interpret, in words, what this confidence interval tells you.
- (e) (3 pts) What is the estimate of the Pearson correlation coefficient in this study? Show your work.

(f) **(6 pts)** The CEO thinks that 6 additional orders will result in no more than 18 additional work hours. Express this statement in terms of a hypothesis test. Does this seem reasonable given the results of this study at the $\alpha = .1$ level? at the $\alpha = .05$ level? Show your work.

(g) **(5 pts)** The company expects 2800 orders next month. This happens to be the same ORDER value as observation #4. Use the output to construct an appropriate 95% interval for the number of work hours next month.

- (h) (4 pts) An assistant presents you with the following plot of the data. Does this pose any problems with the current analysis? Explain your reasoning.



2. Short answer questions. Each part is unrelated.

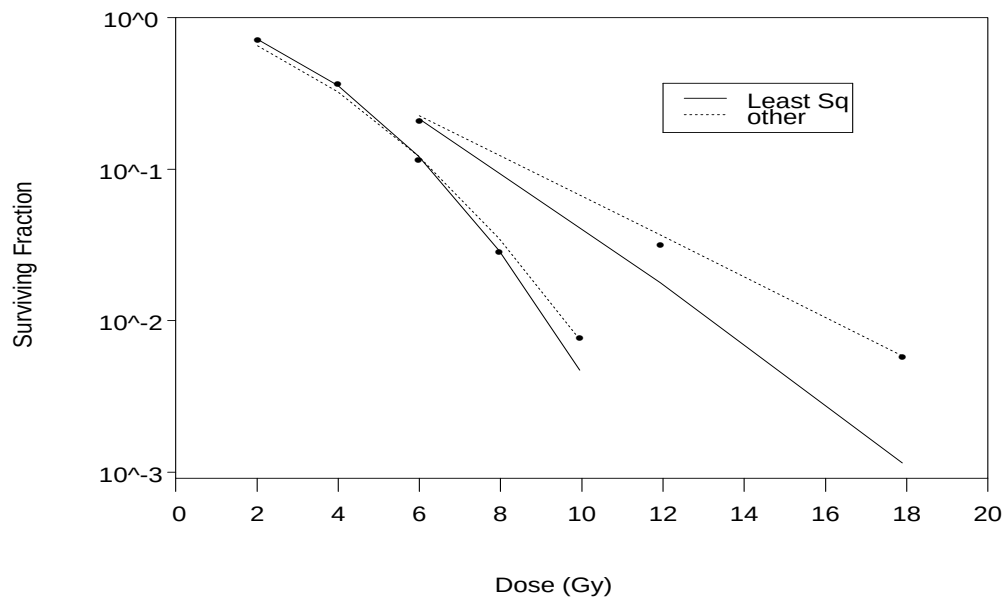
- (a) (4 pts) Explain when, and when not, collinearity among explanatory variables is a problem. When it is a problem, explain how you'd try to remedy the problem.

(b) (9 pts) A new scientist at Purdue comes to you because he has lingering concerns about his analysis. At his previous university, he says a statistician told him to “not worry about the Normality of the response variable because the linear regression procedure is robust.”. He does not understand this explanation and is worried his analysis may not be valid.

- i Please carefully describe to the scientist what is meant by “robust” in terms he can understand.
- ii Please comment on his concern about the Normality of the response variable and how you would try to alleviate this concern. If you think some graphical or numerical summaries would be helpful, please state to the scientist what you’d like him to do before your next meeting.

(c) (7 pts) Consider the general linear test to compare a full model with 4 predictors and a reduced model with only 2 predictors. What is the decision at the $\alpha = .05$ level given that there are $n = 21$ cases, $R^2 = 0.85$ for the full model, and $R^2 = .76$ for the reduced model?

- (d) (4 pts) A client is interested in fitting survival probabilities of a cancer cell for different doses of radiation and two levels of absorption time. He comes to you stating he has developed an alternate way to estimate a regression line that appears better than least-squares. To demonstrate this, he presents you with the following graph that shows his proposed fit and his “least-squares” fit. Give at least one argument why you doubt his claim.



- (e) **(10 pts)** A specific medical device is designed to provide a constant delivery of an anti-inflammatory hormone intravenously. There are three primary brands of this device. You've been asked to help design an experiment to compare them. You have a total of 27 devices (9 from each supplier) and the plan is to look at the relationship between the time the device is on (typically between 1 and 60 minutes) versus the amount of hormone remaining in the device (all devices start with a constant amount). They are interested in your opinions regarding the times to be used for each device and how to go about comparing the relationship between time and amount across the 3 brands. Please provide your recommendations.

(f) (6 pts) A regression analysis resulted in the model

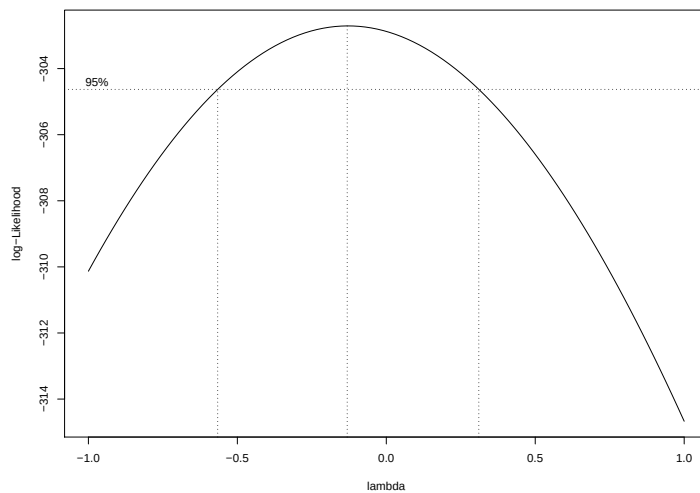
$$\hat{Y}_i = 1.841246 + 0.10934X_i$$

and variance-covariance parameter matrix

$$\begin{bmatrix} 0.1240363 & -0.002627 \\ -0.002627 & 0.0000909 \end{bmatrix}$$

Given that the MSE is 1.6360 for this simple linear model, compute the variance of $\hat{Y}$ when it is used to estimate the mean of the dependent variable when $X = 4$ and compute the variance of $\hat{Y}$ if it is being used to predict a future observation at $X = 4$.

(g) (5 pts) A Box-Cox plot is shown below. Explain how to use this plot and what, if any, actions should be taken for the data in this analysis.



3. A study looked at the violent crime rates (per 100,000 individuals) for 66 different cities in the United States related to three predictors:

- minority- percentage of population not Caucasian
- college- percentage of population ≥ 25 years old with college diploma
- housing- percentage of housing that is not a single family residence

Some regression results are shown below.

Coefficients:

	Estimate	Std. Error	t-value	P-value
(Intercept)	169.3797	8.1292	20.836	0
minority	1.0664	0.1305	8.169	2.02e-11
college	-1.2315	0.2587	-4.760	1.20e-05
housing	0.9185	0.2022	4.543	2.62e-05

Residual standard error, s: 14.94

Overall Model F-statistic : 39.46

- (a) (4 pts) Describe the meaning of the minority and college regression coefficients.

- (b) (6 pts) What are the null and alternative hypotheses and degrees of freedom for i) a coefficient t -test and ii) the overall F test? Please specify the null and alternative hypotheses in words rather than symbols.

(c) (7 pts) Complete the following ANOVA table.

	DF	Type I SS	Mean Sq	F value
minority			17261.5	
college			4563.4	
housing				
error				

(e) (3 pts) What is R^2 for this model?

RESULTS FOR PROBLEM #1

The REG Procedure
Dependent Variable: hours1

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	156854087	156854087	74.02	<.0001
Error	14	29667241	2119089		
Corrected Total	15	186521328			
Root MSE		1455.70899	R-Square	0.8409	
Dependent Mean		7584.50000	Adj R-Sq	0.8296	
Coeff Var		19.19321			

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	514.37418	898.75463	0.57	0.5762
orders	1	2.89315	0.33628	8.60	<.0001

Output Statistics

Obs	Dep Var	Predicted Value	Std Error Mean Predict
1	8065	9194	409.1849
2	10626	10351	485.6408
3	12916	12087	637.4310
4	8907	8615	383.1380
5	7594	6301	393.3326
6	8451	5433	441.5835
7	5888	4565	505.6049
8	2112	4275	529.5026
9	3405	3408	606.7562
10	1554	2250	718.9270
11	2926	4854	482.8677
12	7570	6879	373.0438
13	8875	10062	464.0591
14	12293	10930	532.5638
15	10925	12376	665.3164
16	9245	9772	443.9778